
EDUCATION

Nanyang Technological University, Singapore**Master of Science in Artificial Intelligence****2025 – 2026 (Expected)**

- Relevant Modules: Intro to AI, Deep Neural Networks for NLP, Machine Learning, Deep Learning, Generative AI for Visual Synthesis, Large Language Models, Multi-Agent Systems

Bachelor of Science in Data Science and Artificial Intelligence with Honours (Merit)**2019 – 2023**

- **Awarded NTU Science and Engineering Undergraduate Scholarship** **2019 – 2023**
- Relevant Modules: Data Science, Data Structures, Algorithms, Object Oriented Programming, Software Engineering, Data Analysis, AI, Databases, Data Mining, NLP, Neural Network & Deep Learning, Info Retrieval, Web Development

WORK EXPERIENCE

Singapore General Hospital**Jan 2026 – Present****Research Intern**

- Generated a synthetic ultrasound dataset using OpenCV to address limited labelled medical data.
- Fine-tuned a pre-trained RT-DETR v2 object detection model, achieving $mAP@0.5 = 0.989$ and $mAP@[0.5:0.95] = 0.849$ on a held-out validation set.
- Achieved high precision and recall (~0.98) across multiple detection classes demonstrating strong localization and class separation.

NTU Lee Kong Chian School of Medicine**Sep 2023 – Sep 2024****Research Assistant**

- Co-authored 4+ peer-reviewed publications on infectious disease modelling and simulation.
- Built dynamic lab website, increasing visibility and attracting collaborators from leading research groups.
- Developed statistical models in R to simulate antibody kinetics and population-level immunity dynamics.
- Collaborated across epidemiology, virology, and data science to execute and publish interdisciplinary studies.
- **Publications:**
 1. Singh, A. (2024). *Early neutrophil responses are potential biomarkers to predict severe COVID-19 in adults.*
 2. Ejima, K., Ajelli, M., Singh, A. et al. *Age- and vaccination status-dependent isolation guidelines based on simulation of SARS-CoV-2 Delta cases in Singapore. Commun Med* 5, 76 (2025).
 3. Chua, H.K., Singh, A., et al. (2025). *Defining the Critical Requisites for Accurate Simulation of SARS-CoV-2 Viral Dynamics: Patient Characteristics and Data Collection Protocol.* Journal of Medical Virology, 97, e70174.
 4. Ponce, L., Wang, Y., & Singh, A. (2024). *Modeling antibody kinetics following mRNA booster vaccination and estimating protection durations against SARS-CoV-2 infection.*

Toyota Motor Asia Pacific**Jul 2022 – Dec 2022****NTU EDGE Project**

- Built a PowerBI dashboard to analyze automotive sales and profit metrics across Southeast Asia.
- Automated Excel workflows using VBA, reducing manual work and streamlining monthly reports.

A*Star, Centre for Frontier AI Research**Jun 2022 – Aug 2022****CFAR Internship Award for Research Excellence (CIARE)**

- Contributed to a deep clustering survey under the mentorship of Dr. Pan Yuangang, CFAR.
- Explored unsupervised learning methods applied to industrial-scale datasets.

Apple South Asia Pte. Ltd.**Jun 2021 – Dec 2021****Data Enhancement & Automation Intern**

- Automated business workflows across APAC using Python (xlwings, xlswriter), VBA, Bash, and AppleScript.
- Established intelligent workflows that saved 31.2 business hours per week across the entire team.

PROJECTS

SentiStock**Nov 2025**

- Designed a modular NLP pipeline integrating financial-aware preprocessing, entity-aware ranking, and semantic similarity expansion to prioritize high-impact market news.
- Fine-tuned a Flan-T5-Base model on synthetically generated financial news, achieving 95.85% sentiment classification accuracy while generating interpretable sentiment rationales for investment decision support.

VehicleRemixStudio**Nov 2025**

- Built an end-to-end, DiffEdit-inspired controllable image editing system that allows users to modify vehicles and backgrounds via text prompts while preserving object structure and scene coherence.

- Implemented the full pipeline end-to-end, designing a U-Net segmentation model from scratch, integrating it with Stable Diffusion inpainting, and wiring it into an interactive UI for prompt-driven visual editing.

London House Price Prediction Kaggle Competition

Nov 2025

- Independently achieved a Top 5.9% leaderboard rank (30 / 511) in a Kaggle-style London House Price Prediction competition by developing a robust tabular regression pipeline.
- Built and manually optimized a single-model XGBoost regressor (after evaluating LightGBM, CatBoost, and ensembles), achieving cross-validated MAE of £138,777 through extensive feature engineering, 5-fold CV, and aggressive hyperparameter tuning.

OmLine

Jun 2025

- Built an AI-powered, real – time yoga posture coach using OpenCV and MediaPipe; trained a classifier on joint-angle features extracted from pose images.
- Delivered real-time multimodal feedback with skeletal overlays and voice corrections, enhancing accessibility in remote fitness.

Final Year Project

Aug 2022 – May 2023

Learning Conditional Human Motion Prior

- Developed a system that enabled natural language control of 3D virtual avatars, generating coherent and meaningful human-like movements suited to specific scenarios using Brax.
- By leveraging a conditional human motion prior that considered scene information and language descriptions, and combining language-conditioned human motion generation and physics-based character control techniques, the team achieved their objective of producing physics-aware virtual characters from textual inputs.
- **Singh, A. (2023).** *Animate your avatar: Learning conditional human motion prior* (Final Year Project, Nanyang Technological University, Singapore).

Age, Gender and Ethnicity Classification Model

Sep 2022

- Developed and trained CNN models, including Ground-up CNN, Inception V3, and VGG16, to predict gender, age, and ethnicity from facial data for security purposes, achieving 95.3% accuracy using a bagged model approach.

Machine Learning Project

Apr 2022

- Developed a solution to recognize blood-related faces leveraging techniques such as Facial Recognition, DeepFace, Classification, and Ensemble Learning.

Software Engineering Project

Feb – Apr 2021

- Developed and integrated the UI for a job and course recommendation software using React JS, contributed to frontend-backend integration, and adhered to the SDLC for seamless development, testing, and deployment.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

Build For Impact Hackathon

Jan 2026

- Built Cycle.it, a circular-economy platform enabling users to rent, swap, thrift, or donate pre-owned items to promote sustainable consumption

NAE Nanyang Technological University

2020 – 2023

President of Nanyang Arts Ensemble

2021 – 2022

- Led a team of 25 members to organise and host arts and crafts workshops for NTU students, demonstrating strong leadership skills in managing and coordinating the team.
- Collaborated with other organizations and CCAs to improve publicity and outreach, displaying effective communication and teamwork skills.

Won NTU IEEE iNTUition Hackathon in the best women in tech category

28 Feb 2021

Operations and Events Director in NTU Model United Nations

2020 – 2021

Publicity Officer at MLDA @ EEE

2020 – 2021

SKILLS & INTERESTS

Technical Skills

Python, Java, JavaScript, HTML5, CSS3, Tailwind, C, R, SQL, Bash, VBA, MATLAB, Computer Vision, Natural Language Processing, Diffusion Models, Object Detection, Tabular Machine Learning, Fine-tuning, Domain Adaptation, Synthetic Data Generation, Cross-Validation, Hyperparameter Tuning, Feature Engineering

Frameworks & Libraries

TensorFlow, PyTorch, Keras, React.js, MediaPipe, DeepFace

Tools & Platforms

PowerBI, Blender, Git, MonolixSuite, MongoDB

Online Courses

Introduction to JavaScript, Beginner in NodeJS, Flutter, Big Data, Apache Hadoop, Apache Spark, Data Management, MongoDB, Data Modelling

Languages

English (Fluent), Hindi (Native), French (Beginner)